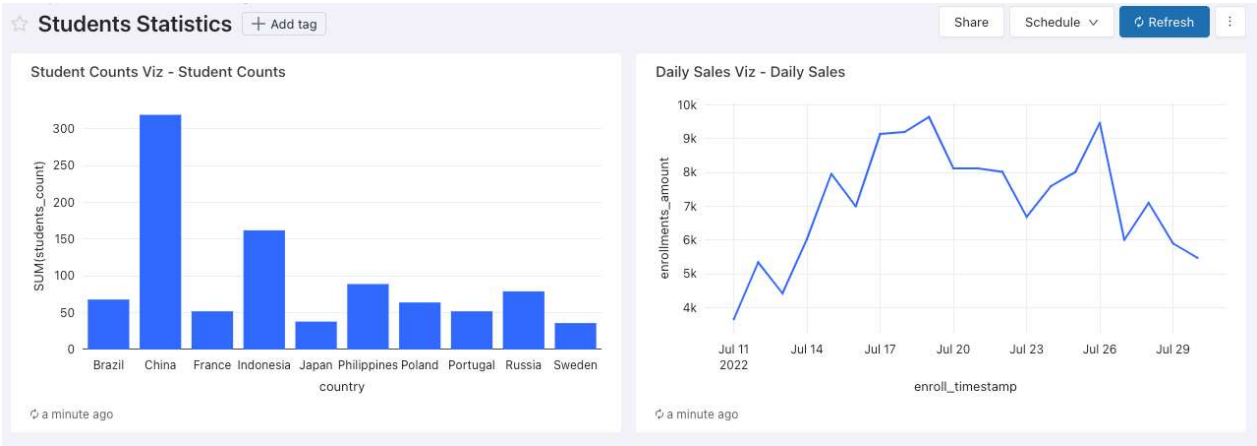
(https://databricks.com)

Lab Solution: Design a Dashboard with DBSQL

In this lab, we will design a dashboard in DBSQL that has 2 graphs:

1. Bar graph that shows the number of students per country
2. Line graph that shows the daily enrollments amount

- Help: Databricks SQL documentation (https://docs.databricks.com/sql/index.html#what-is-databricks-sql).



Q1 - Working with queries in SQL Editor

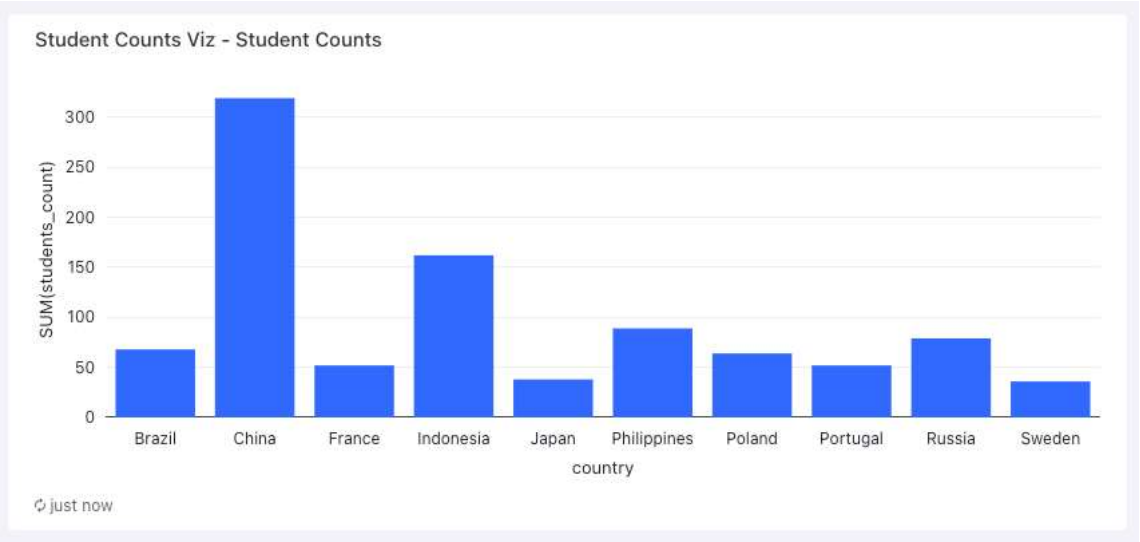
Run the the below query in **SQL Editor** in Databricks SQL, and then save it with the name **Student Counts**

5

```
SELECT profile:address:country as country, count(student_id) AS students_count
FROM hive_metastore.de_associate_school.students
GROUP BY profile:address:country
ORDER BY students_count DESC
LIMIT 10
```

Q2 - Creating a Bar Graph Visualization

Create a bar graph that shows the number of students per country



Anwser:

Steps:

Q3 - Creating a New Dashboard

Add the above graph to a new dashboard named **Students Statistics**

Anwser:

Steps:

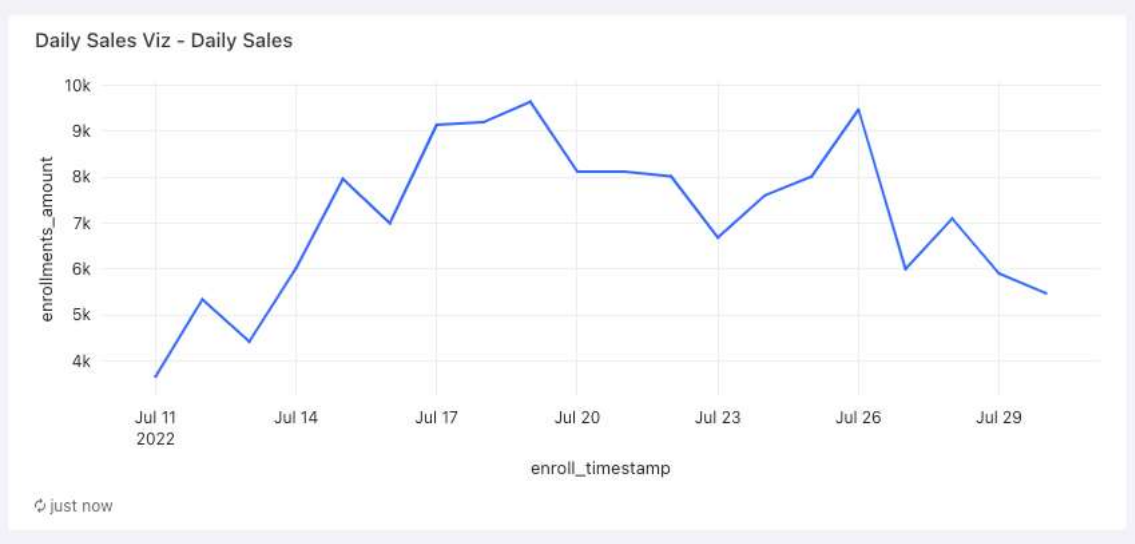
1. Click the three vertical dots button at the top of the graph and select **Add to Dashboard**.
2. Click the **Create new dashboard** option
3. Name your dashboard **Students Statistics**
4. Click **Save**
5. With the new dashboard selected as the target, click **OK** to add your visualization

Q4 - Creating a Line Plot Visualization

1. Run the the below query in a new query tab in the **SQL Editor**, and then save it with the name **Daily Sales**


```
SELECT cast(from_unixtime(enroll_timestamp, 'yyyy-MM-dd HH:mm:ss') AS date) enroll_timestamp,
       sum(total) AS enrollments_amount
FROM hive_metastore.de_associate_school.enrollments n
INNER JOIN hive_metastore.de_associate_school.students s ON s.student_id = n.student_id
GROUP BY enroll_timestamp
```

2. Create a Line Plot Visualization that shows the daily enrollments amount



Anwser:

Steps:

1. Click the **Add Visualization** button
2. Select `Line` for the **Visualization Type**
3. Set `enroll timestamp` for the **X Column**

Q5 - Review your Dashboard

Open your Dashboard and refresh its underlaying data

Anwser:

Steps:

1. Click on the **Dashboards** button on left side bar
2. Find the dashboard **Students Statistics** created earlier. Click to open it
3. Click the **Refresh** button to update your dashboard